

Generalized Switch-Setting Problems

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Abstract

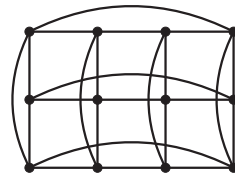
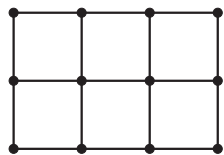
Switch-setting games like Lights Out are typically modelled as a graph, where the vertices represent switches and lamps, and the edges capture the switching rules. We generalize the concepts used for a mathematical description of Lights Out and its relatives. Our approach uses bipartite graphs and allows for the analysis of a broader class of switch-setting games, which is demonstrated at a new variant of the Lights Out puzzle. Our method exhibits full duality between switches and lamps, and we get rid of some insufficiencies inherent in the modelling with non-bipartite graphs. We present a detailed analysis of the new Lights Out variant, formulate solvability conditions, give graphically aesthetic interpretations and discuss aspects on minimum solutions. In our study of parity domination in bipartite graphs we incorporate methods from linear algebra and linear programming. We point out the close relations between graph theoretic terms and the language of algebra over $\mathbb{Z}_2$.

Key words: Lights Out, Merlin's Magic Square, switch-setting game, parity domination, bipartite graph

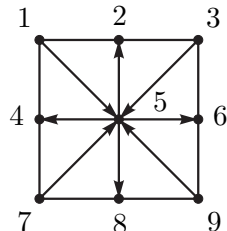
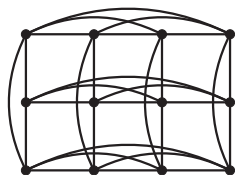
1 Introduction

Lights Out [1] is a one-person game played on a 5×5 board of lamps, each of which is in either on- or off-state. Toggling one lamp also causes the rectilinear adjacent lamps to change state. The challenge is to transform an initial light pattern into the all-lamps-off pattern by a sequence of switching operations. An initial light pattern is said to be *solvable*, if such a sequence of switching operations exists.

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(a) Lights Out on a 3×4 grid of lamps. (b) Lights Out on a 3×4 torus.



(c) Araújo's [2] Lights Out variant. (d) Merlin's Magic Square.

Fig. 1. Four switch-setting games (closed neighborhood switching operations).

Lights Out can be modelled as a 5×5 grid graph¹, with a binary state assigned to each vertex. Inverting the state of one vertex also inverts the states of the adjacent vertices in the grid. Instead of a 5×5 grid, the game can be played more generally on an $m \times n$ grid graph or on a torus (Fig. 1 (a) and (b) for $m = 3$, $n = 4$). Araújo [2] discusses a variant played on an $m \times n$ board, where toggling one lamp causes all lamps on the same row and all lamps on the same column to change state (Fig. 1 (c) for $m = 3$, $n = 4$). Another switch-setting game is Merlin's Magic Square [3,4]; it is played on the non-symmetric graph from Fig. 1 (d).

For the variants discussed up to here, a switching operation changed the state of all vertices in the *closed* neighborhood of a given vertex (cp. Sutner's σ^+ -rule in the language of cellular automata [12]). If instead we define a switching operation to change the state of all vertices in the *open* neighborhood (cp. Sutner's σ -rule), we have to add self-loops at all the vertices from the graphs in Fig. 1 to capture the same games (cp. Fig. 1 (d) and Fig. 3 (a)). Orbix is a switch-setting game based on open neighborhood switching operations; it is played on the regular icosahedron.

¹ Throughout this article we consider finite directed graphs. We denote with V the set of vertices and with $E \subseteq V \times V$ the set of edges. By convention, two directed edges $(u, v), (v, u) \in E$ are commonly interpreted as a single undirected edge. The *open* and *closed neighborhood* of a vertex $u \in V$ are defined by $N(u) := \{v \in V \mid (u, v) \in E\}$ and $\bar{N}(u) := N(u) \cup \{u\}$, respectively. A graph is said to be (non-)reflexive or (non-)symmetric, if E is (non-)reflexive or (non-)symmetric, respectively. The *adjacency matrix* $A : V \times V \rightarrow \{0, 1\}$ of a graph is defined by $A(u, v) = 1 \iff (u, v) \in E$.

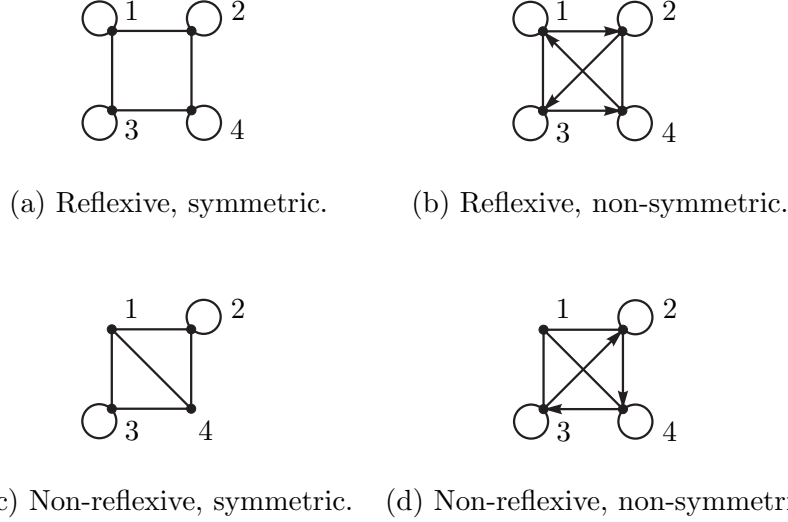
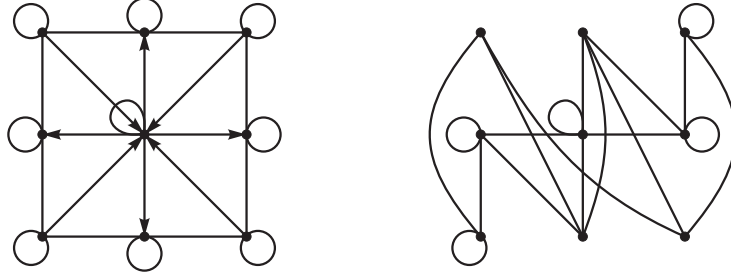


Fig. 2. Different underlying graphs equivalently representing Lights Out on a 2×2 grid (open neighborhood switching operations).

The strategies to solve switch-setting problems range across different branches of mathematics. Methods from linear algebra study the existence and uniqueness of solutions for certain systems of linear equations over $\mathbb{Z}_2$ [1–3, 5, 8, 9], a graph theoretical approach leads to parity domination problems (see [7] and references therein), whereas in the language of cellular automata the solvability situation translates into reversibility properties of certain additive cellular automata [11–15]. Gaining insight into the relations between these areas makes switch-setting problems a fascinating study object, and in this article we will occasionally refer to alternative formulations.

To derive propositions about the solvability of certain initial light patterns in a switch-setting game, commonly reflexivity and symmetry of the underlying graphs are exploited [8, 9, 11]. These properties, however, do not necessarily capture the essence of solvability. Fig. 2 (a) shows the underlying graph for Lights Out played on a 2×2 grid. It is reflexive and symmetric. The graphs depicted in Fig. 2 exhibit all possible combinations of these two properties. Surprisingly, all four graphs show the same behavior regarding which initial light patterns are solvable and which are not. The subsets of the vertex set $V = \{1, 2, 3, 4\}$ defined by the neighborhood relation are the same for each of the four graphs; more precisely, the set $\{N(v) \mid v \in V\}$ is always the same. To put it another way, the adjacency matrices of these four graphs differ only in the order of their rows (Eriksson et al. discuss a different algebraic procedure to obtain equivalent switch-setting games [6, Th. 4.1]). Appropriately permuting the rows of the adjacency matrix of the reflexive, non-symmetric graph from Fig. 3 (a) yields the adjacency matrix of the non-reflexive, symmetric graph from Fig. 3 (b) that equivalently represents the solution behavior of Merlin’s Magic Square.



(a) Reflexive, non-symmetric. (b) Non-reflexive, symmetric.

Fig. 3. Different underlying graphs equivalently representing Merlin's Magic Square (open neighborhood switching operations).

We introduce a new Lights Out variant, played on an $m \times n$ board of lamps. A switching operation inverts the states of all lamps on either a row, a column, a diagonal or an antidiagonal. All precedingly mentioned switch-setting games identified switches and lamps, in particular the number of switches was equal to the number of lamps. Our Lights Out variant is fundamentally different in this regard, since $m + n + 2(m + n - 1) = 3(m + n) - 2$ switches and mn lamps have to be considered.

In Section 2 we incorporate methods from graph theory and linear algebra to generalize the concepts used for a mathematical description of switch-setting games. We introduce the notion of a *generalized switch-setting problem* that allows for an analysis of the new Lights Out variant and fully dual observations between switches and lamps. The difficulties, whether a switching operation acts on the open or the closed neighborhood of a vertex, disappear, as well as problems with different graphs describing essentially the same game. With these terms at hand we discuss parity domination in bipartite graphs and solvability conditions for generalized switch-setting problems, and we recapitulate Sutner's odd-parity cover theorem [11].

This mathematical machinery is applied to our new Lights Out variant in Section 3. We present a detailed analysis of the game and some nice graphical interpretations. We thereby solve the following geometrical problem that shall be stated here to illustrate the aesthetic dimension of the matter. The solution is given at the end of Section 3.

Problem *Consider an infinite chessboard. Over each row, column, diagonal or antidiagonal of squares a straight line through the centers of the respective squares may be drawn. Characterize line patterns that cause an even number of lines to intersect in each square!*

In Section 4 we discuss aspects on minimum solutions for our new Lights Out variant, where surprisingly a linear programming problem appears.

2 Generalized Switch-Setting Problems

Let us fix some notation that will be used throughout the article. All arithmetic is performed over the field $\mathbb{Z}_2$. Let U be a finite set. We identify a vector $x : U \rightarrow \{0, 1\}$ and a subset of U by means of the bijection $\phi_U : \{0, 1\}^U \rightarrow \mathfrak{P}(U)$, defined by $u \in \phi_U(x) :\iff x(u) = 1$. Let I and K be finite sets and $I' \subseteq I$, $K' \subseteq K$. As a notational convenience, for a matrix $A : I \times K \rightarrow \{0, 1\}$ we denote by $A_{I', K'}$ the restriction $A|(I' \times K')$.

Definition 1 *The pair (G, z_0) is an ordinary switch-setting problem if G is a graph with vertex set V and $z_0 : V \rightarrow \{0, 1\}$.*

In this formalization of Lights Out and its relatives, switches and lamps are modelled by the vertices of a graph whose edges represent the switching rules.

Let (G, z_0) be an ordinary switch-setting problem, V the vertex set and A the adjacency matrix of G . The light pattern as a result of toggling a switch $v \in V$ can be calculated as $z_0 + A^T \chi_v \pmod{2}$, where $\chi_v : V \rightarrow \{0, 1\}$ is defined by $\chi_v(u) = 1 :\iff u = v$. We search a sequence of switches $v_0, v_1, \dots, v_k \in V$, such that

$$z_0 + A^T \chi_{v_0} + A^T \chi_{v_1} + \dots + A^T \chi_{v_k} = z_0 + A^T \underbrace{(\chi_{v_0} + \chi_{v_1} + \dots + \chi_{v_k})}_{=:x} = 0 \pmod{2}$$

holds. The solution behavior of the ordinary switch-setting problem is therefore fully captured by the system of equations

$$A^T x = z_0 \pmod{2} \tag{1}$$

with $x \in \{0, 1\}^V$.

Note, that transforming an initial state z_0 into a state z_1 requires the same switching operations as transforming $z_0 + z_1$ into the zero state.

Definition 2 *The 4-tuple (G, S, L, z_0) is a generalized switch-setting problem if G is a symmetric, bipartite graph with an induced partition of its vertices into the sets S and L and $z_0 : L \rightarrow \{0, 1\}$.*

Switches and lamps are modelled as the two component vertex sets (S and L) of a bipartite graph. The number of switches and the number of lamps are therefore independent from one another. Since no state information is assigned to the switches, a switching operation naturally changes the state of all lamps in the *open* neighborhood of a given switch vertex.

Let (G, S, L, z_0) be a generalized switch-setting problem and A the adjacency matrix of G . The solution behavior of the generalized switch-setting problem

is fully captured by the system of equations

$$A_{S,L}^T x = A_{L,S} x = z_0 \pmod{2} \quad (2)$$

with $x \in \{0, 1\}^S$.

Conversely, any system of linear equations $Mx = z$ over $\mathbb{Z}_2$ can be solved experimentally by considering a generalized switch-setting problem (G, S, L, z) , where the graph G is constructed such that its adjacency matrix A satisfies the condition $A_{L,S} = M$.

The following proposition gives a construction to expand an arbitrary graph to a symmetric, bipartite graph while preserving the adjacency relation. It constitutes the foundation for translating every ordinary switch-setting problem into a generalized one. The proposition is stated without a proof.

Proposition 3 *Let $G' = (V', E')$ be a graph and A' its adjacency matrix. Then $G = (V, E)$, defined by*

$$\begin{aligned} S &:= V' \times \{0\} \\ L &:= V' \times \{1\} \\ V &:= L \cup S \\ ((u', 0), (v', 1)), ((v', 1), (u', 0)) &\in E \iff (u', v') \in E' \end{aligned}$$

is a symmetric, bipartite graph with an induced partition of V into the sets S and L , and the restriction of its adjacency matrix A on $S \times L$ differs from A' only in the naming of rows and columns. More precisely, for all $u', v' \in V'$

$$A'(u', v') = A_{S,L}((u', 0), (v', 1)).$$

Applying this construction to each graph from Fig. 2 we obtain bipartite graphs that are easily seen to be isomorphic (the construction is illustrated in Fig. 4). The solution behavior of switch-setting problems based on these graphs is therefore identical. The same holds for the graphs in Fig. 3.

The inverse construction, flattening a symmetric, bipartite graph while preserving the adjacency relation, is described in the following proposition that is also stated without a proof. It allows a subset of generalized switch-setting problems to be written as ordinary ones.

Proposition 4 *Let $G = (V, E)$ be a symmetric, bipartite graph with an induced partition of its vertices into sets S and L with $|S| = |L|$ and adjacency matrix A . Let $\tau : S \rightarrow L$ be an arbitrary bijection.*

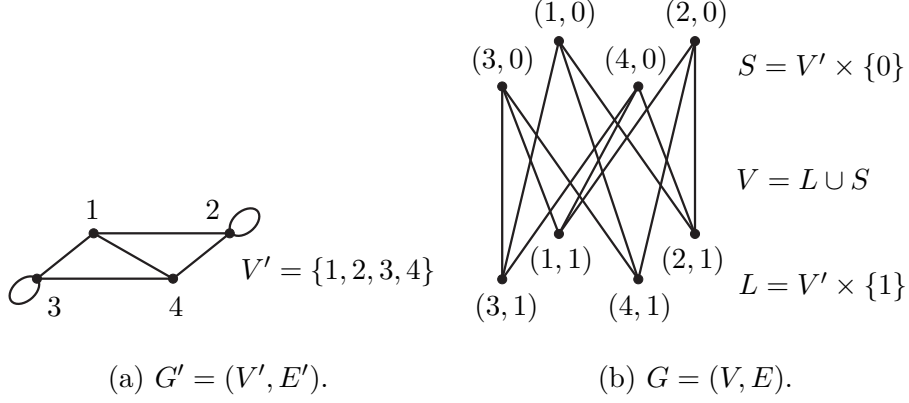


Fig. 4. Construction from Proposition 3 with G' from Fig. 2 (c).

Then $G' = (V', E')$, defined by

$$V' := L$$

$$(l_1, l_2) \in E' \iff \bigvee_{s \in S} (\tau(s) = l_1 \wedge (s, l_2) \in E)$$

is a graph, whose adjacency matrix A' differs from $A_{S,L}$ only in the naming of rows. More precisely, for all $s \in S$ and all $l \in L$

$$A_{S,L}(s, l) = A'(\tau(s), l).$$

The freedom in choosing a bijection τ can be used to obtain different flattened graphs. Since their adjacency matrices differ only in the naming of rows, the solution behavior of switch-setting problems based on these graphs is the same.

Definition 5 Let $G = (V, E)$ be a graph.

A subset $X \subseteq V$ is said to be an even dominating set or odd dominating set of G if for all $v \in V$ the set $N(v) \cap X$ is of even cardinality or odd cardinality, respectively.

Definition 6 Let $G = (V, E)$ be a bipartite graph with an induced partition of its vertices into the sets V_0 and V_1 .

A subset $X \subseteq V_0$ is said to be an even V_1 -dominating set or odd V_1 -dominating set of G if for all $v_1 \in V_1$ the set $N(v_1) \cap X$ is of even cardinality or odd cardinality, respectively.

Given a generalized switch-setting problem (G, S, L, z_0) , toggling all switches in an even L -dominating set of G does not change the state of any lamp, while toggling all switches in an odd L -dominating set of G inverts the state of every lamp. Observe, that the empty set is an even L -dominating set and an even S -dominating set.

The following proposition relates the algebraic description of a generalized switch-setting problem (2) with the even dominating sets of the underlying graph.

Proposition 7 *Let $G = (V, E)$ be a symmetric, bipartite graph with an induced partition of its vertices into the sets S and L and adjacency matrix A . Then it is*

$$\begin{aligned} x \in \ker(A_{L,S}) &\overset{X=\phi_S(x)}{\iff} X \subseteq S \text{ is an even } L\text{-dominating set of } G \\ z \in \ker(A_{S,L}) &\overset{Z=\phi_L(z)}{\iff} Z \subseteq L \text{ is an even } S\text{-dominating set of } G. \end{aligned}$$

PROOF. This follows from (2) with the definition of the kernel of a linear mapping and the definition of an even dominating set. $\square$

Theorem 8 *Let $P = (G, S, L, z_0)$ be a generalized switch-setting problem and A the adjacency matrix of G .*

The following statements are equivalent:

- (i) P has a solution.
- (ii) The number of lit lamps on every even S -dominating set of G is even.
- (iii) The number of lit lamps on every even S -dominating set from a basis of the set of all even S -dominating sets of G is even.

PROOF. Consider the following auxiliary statements:

- (i') $A_{L,S} x = z_0 \pmod{2}$ has a solution.
- (ii') For all $z \in \ker(A_{S,L})$ the relation $z^T z_0 = 0 \pmod{2}$ holds.
- (iii') For all z from a basis of $\ker(A_{S,L})$ the relation $z^T z_0 = 0 \pmod{2}$ holds.

The equivalence between (i) and (i') follows from (2). The equivalence between (i'), (ii') and (iii') follows from linear algebra (see e.g. [16, p. 81] for proof). The equivalence between (ii) and (ii') follows from Proposition 7, as well as the equivalence between (iii) and (iii'). $\square$

With the last statement Theorem 8 establishes a necessary and sufficient condition for the solvability of a generalized switch-setting problem, that is easy to check, once a basis for the even S -dominating sets of the underlying graph has been determined. If a solution x of (2) is known, the set of all solutions is given by $x + \ker(A_{L,S})$. Since the kernel of $A_{L,S}$ corresponds to the even L -dominating sets of the underlying graph, together with the even S -dominating sets they play the major role in examining the solution behavior of a switch-setting problem. The roles of switches and lamps interchange, if instead of

$P = (G, S, L, z_0)$ the problem $P' = (G, L, S, x_0)$ with an $x_0 \in \{0, 1\}^S$ is considered. This fact underlines the fully dual behavior of switches and lamps introduced by the concept of generalized switch-setting problems. This duality is inherited from the duality between the column and the row space of $A_{L,S}$ and between its right and left null space. The generalized switch-setting problem provides therefore an illuminating illustration for the four fundamental subspaces of a matrix. This clarity becomes blurred for the ordinary switch-setting problem on a symmetric graph, where column and row space of the adjacency matrix are identical, as well as its right and left null space.

In the following we apply Theorem 8 to special cases. First, we examine under which conditions all possible initial light patterns on the same graph are solvable. The proof of the following corollary is left to the reader.

Corollary 9 *Let G be a symmetric, bipartite graph with an induced partition of its vertices into the sets S and L , and $\mathcal{P} = \{(G, S, L, z) \mid z \in \{0, 1\}^L\}$ the set of generalized switch-setting problems with underlying graph G . The following statements are equivalent:*

- (i) *Every problem $P \in \mathcal{P}$ has a solution.*
- (ii) *For each lamp there exists a sequence of switching operations that results in the inversion of this lamp's state only.*
- (iii) *The empty set is the only even S -dominating set of G .*

Let us briefly relate this result to alternative formulations in the case of ordinary switch-setting problems. Eriksson et al. give an interesting characterization for the solvability of all possible ordinary switch-setting problems on the same underlying graph in terms of the number of complete matchings in the graph [6, Th. 6.1]. In the language of cellular automata, solvability of all possible initial light patterns is equivalent to the reversibility of the corresponding σ^- or σ^+ -automaton [12, p. 27].

By determining sets of switches that invert the state of individual lamps only we see that every initial light pattern for Merlin's Magic Square is solvable. Because of the symmetry of the underlying graph (see Fig. 1 (d)) it suffices to find sets of switches that invert the state of a corner lamp, a lamp along an edge or the central lamp, such sets are given by $\{3, 5, 6, 7, 8, 9\}$, $\{1, 2, 6, 7, 8\}$ and $\{2, 4, 5, 6, 8\}$, respectively.

We ask now under which conditions the initial all-lights-on pattern is solvable. Again, the proof of the following corollary is omitted.

Corollary 10 *Let $P = (G, S, L, z_0)$ be a generalized switch-setting problem with $z_0(l) = 1$ for all $l \in L$. The following statements are equivalent:*

- (i) P has a solution.
- (ii) There exists an odd L -dominating set of G .
- (iii) Every even S -dominating set of G has an even number of elements.

How parity domination in graphs, that are not necessarily bipartite, fits into this framework, is surveyed briefly in the last part of this section.

Proposition 11 *Let $G = (V, E)$ be a reflexive, symmetric graph. Then every even dominating set of G has an even number of elements.*

PROOF. Let $<$ denote an arbitrary total strict order on V and A the adjacency matrix of G . For an even dominating set X of G , $x := \phi_V^{-1}(X)$ is an element of $\ker(A)$, i.e. $Ax = 0$ holds. From

$$0 = Ax = \underbrace{x^T Ax}_{=0} = x^T x + 2 \sum_{\substack{u, v \in V \\ u < v}} x(u)x(v)A(u, v) = x^T x = \sum_{v \in V} x(v) \pmod{2}$$

the proposition follows (cp. [8]). $\square$

Theorem 12 (Sutner [11]) *Every reflexive, symmetric graph has an odd dominating set.*

PROOF. From Proposition 11 we obtain that every even dominating set of the graph has an even number of elements. Restating Corollary 10 analogously for ordinary switch-setting problems based on a symmetric graph eventually yields the statement. $\square$

Therefore, an ordinary switch-setting problem based on a reflexive, symmetric graph with the initial all-lights-on pattern is solvable. Theorem 12 can be generalized to show, that if the graph is assumed to be symmetric, but not necessarily reflexive, the initial light pattern with lit lamps on all self-loop vertices is solvable. This follows from the fact that (1) has a solution for a symmetric matrix A and $z_0 = \text{diag}(A^T) = \text{diag}(A)$, where $\text{diag}(A)$ denotes the main diagonal of A . Lossers [8] gives an algebraic proof for that, Dodis and Winkler [9, Lemma 3.2] use a double counting argument, whereas Chen [10] proposes an inventive purely graph theoretic construction.

3 Circles and Tilings

With the tools from Section 2 at hand we examine a new Lights Out variant, played on an $m \times n$ board of lamps. A switching operation inverts the states of all lamps on either a row, a column, a diagonal or an antidiagonal. In particular each of the four lamps in a board corner can be toggled independently from any other lamps. For this game the ordinary switch-setting problem is not the appropriate model, since the number of switches $m + n + 2(m + n - 1) = 3(m + n) - 2$ is in general different from the number of lamps mn . Hence, we model our Lights Out variant as a generalized switch-setting problem with the underlying symmetric, bipartite graph $G_{m,n} = (V, E)$ defined by

$$\begin{aligned} S &:= \{s_{\rightarrow 1}, s_{\rightarrow 2}, \dots, s_{\rightarrow m}\} \cup \{s_{\downarrow 1}, s_{\downarrow 2}, \dots, s_{\downarrow n}\} \\ &\quad \cup \{s_{\searrow 1-m}, s_{\searrow 2-m}, \dots, s_{\searrow n-1}\} \cup \{s_{\swarrow 1-m}, s_{\swarrow 2-m}, \dots, s_{\swarrow n-1}\} \\ L &:= \{l_{i,j} \mid i = 1, 2, \dots, m; j = 1, 2, \dots, n\} \\ V &:= S \cup L \end{aligned}$$

and

$$(s, l_{i,j}), (l_{i,j}, s) \in E : \Longleftrightarrow s = s_{\rightarrow i} \vee s = s_{\downarrow j} \vee s = s_{\searrow j-i} \vee s = s_{\swarrow n-i-j+1}.$$

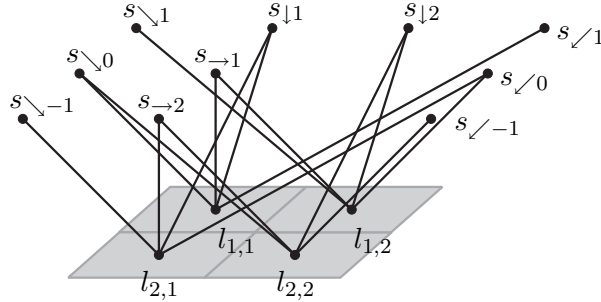


Fig. 5. The bipartite graph $G_{2,2}$.

Let A denote the adjacency matrix of $G_{m,n}$. Throughout this and the next section $G_{m,n}$, V , E , S , L and A are used as standing notations with respect to this definition, parametrized by $m, n \in \mathbb{N}_{>0}$. Note, that we will frequently apply Proposition 7 and make use of the immediate correspondence between the even S -dominating sets of $G_{m,n}$ and $\ker(A_{S,L})$ and between the even L -dominating sets and $\ker(A_{L,S})$.

To exploit the solution behavior of our game we follow the outline from Section 2 and determine the even S -dominating sets and the even L -dominating sets of $G_{m,n}$.

Observe, that an even S -dominating set of $G_{m,n}$ has an even number of elements in common with every row, column, diagonal and antidiagonal of lamps,

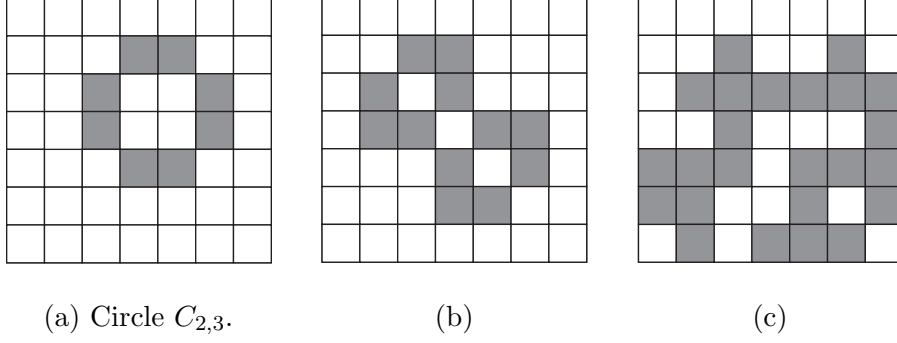


Fig. 6. Three even S -dominating sets of $G_{7,7}$. Shown is the board of lamps with the elements of an even S -dominating set colored grey.

and is therefore of even cardinality (Fig. 6).

Proposition 13 *Let $\min(m, n) < 4$, then the empty set is the only even S -dominating set of $G_{m,n}$.*

PROOF. The case $\min(m, n) = 1$ is trivial. Without any loss of generality we now assume $1 < m < 4$. Suppose, there was a nonempty even S -dominating set Z . Let R be the smallest rectangular subboard such that $R \supseteq Z$. Consider the lamps on the rightmost column of R . Two of them are elements of Z , at least one of which lies in a corner of R . Hence, there exists a switch for a diagonal or an antidiagonal of lamps with only one lamp from Z in its neighborhood, i.e. an odd number. This is a contradiction, since Z is an even S -dominating set. $\square$

For $\min(m, n) \geq 4$ the graph $G_{m,n}$ has nonempty even S -dominating sets (Fig. 6). In search of a basis for these sets we observe that for $i, j, d \in \mathbb{N}$ with $1 \leq i \leq m$ and $1 \leq j < j + d \leq n$

$$\{l_{i,j}, l_{i,j+1}\} \triangle \{l_{i,j+1}, l_{i,j+2}\} \triangle \cdots \triangle \{l_{i,j+d-1}, l_{i,j+d}\} = \{l_{i,j}, l_{i,j+d}\}.$$

Similarly, for $i, j, d \in \mathbb{N}$ with $1 \leq i < i + d \leq m$ and $1 \leq j < j + d \leq n$

$$\{l_{i,j}, l_{i+1,j+1}\} \triangle \{l_{i+1,j+1}, l_{i+2,j+2}\} \triangle \cdots \triangle \{l_{i+d-1,j+d-1}, l_{i+d,j+d}\} = \{l_{i,j}, l_{i+d,j+d}\}.$$

This motivates the following

Definition 14 *A circle $C_{i,j}$ is a subset of L , defined for $i \in \{1, 2, \dots, m-3\}$ and $j \in \{1, 2, \dots, n-3\}$ by*

$$C_{i,j} := \{l_{i,j+1}, l_{i,j+2}, l_{i+1,j}, l_{i+1,j+3}, l_{i+2,j}, l_{i+2,j+3}, l_{i+3,j+1}, l_{i+3,j+2}\}.$$

Theorem 15 *The set of circles is a basis for the set of even S -dominating sets of $G_{m,n}$. The following algorithm gives for any even S -dominating set its representation as a linear combination of circles.*

Algorithm

Input. An even S -dominating set Z of $G_{m,n}$.

Output. A set of lamps $L' \subseteq L$ such that $Z = \Delta_{l_{i,j} \in L'} C_{i,j}$ holds, where we denote with $\Delta_{l_{i,j} \in L'} C_{i,j}$ the symmetric set difference between all circles $C_{i,j}$ with $l_{i,j} \in L'$.

1. Put $L' := \emptyset$ and $Z' := Z$.
2. Determine the smallest rectangular subboard R such that $R \supseteq Z'$.
3. Let i denote the row index of the uppermost row of R and 2λ the number of lamps from Z' in this row. Partition the lamps of Z' in row i into a family of sets $(\{l_{i,j_k}, l_{i,j'_k}\})_{k=1,\dots,\lambda}$ such that $j_1 < j'_1 < j_2 < j'_2 < \dots < j_\lambda < j'_\lambda$ holds.
4. Put

$$L' := L' \cup \bigcup_{k=1}^{\lambda} \{l_{i,j_k-1}, l_{i,j_k}, l_{i,j_k+1}, \dots, l_{i,j'_k-3}, l_{i,j'_k-2}\}$$

$$Z' := Z \Delta \bigtriangleup_{l_{i,j} \in L'} C_{i,j}.$$

5. If $Z' = \emptyset$ go to step 6, otherwise go to step 2.
6. The algorithm terminates, the relation $Z = \Delta_{l_{i,j} \in L'} C_{i,j}$ holds.

PROOF. A circle is an even S -dominating set. Each set of circles is linearly independent, i.e. no circle is the symmetric set difference of other circles.

Step 3: This partition is possible, since the number of lamps from Z' in row i is even (Z' is an even S -dominating set).

Step 4: The lamps $l_{i,1}$ and $l_{i,n}$ are not elements of Z' , otherwise there was a switch for a diagonal or an antidiagonal of lamps, having from Z' only $l_{i,1}$ or $l_{i,n}$ in its neighborhood. Therefore it is $1 \leq j_1 - 1$ and $j'_\lambda - 2 \leq n - 3$, and the expression $\Delta_{l_{i,j} \in L'} C_{i,j}$ is well defined. The even S -dominating sets are closed under the symmetric set difference operator and Z' is an even S -dominating set again (recall Proposition 7 and that the kernel of a linear mapping is a vector space).

Steps 5 and 6: By construction of the set Z' , the height of the rectangular subboard R decreases in every run at least by 1. According to Proposition 13 the algorithm terminates after at most $m - 3$ runs, and from $Z' = Z \Delta \bigtriangleup_{l_{i,j} \in L'} C_{i,j} = \emptyset$ we eventually obtain $Z = \Delta_{l_{i,j} \in L'} C_{i,j}$. $\square$

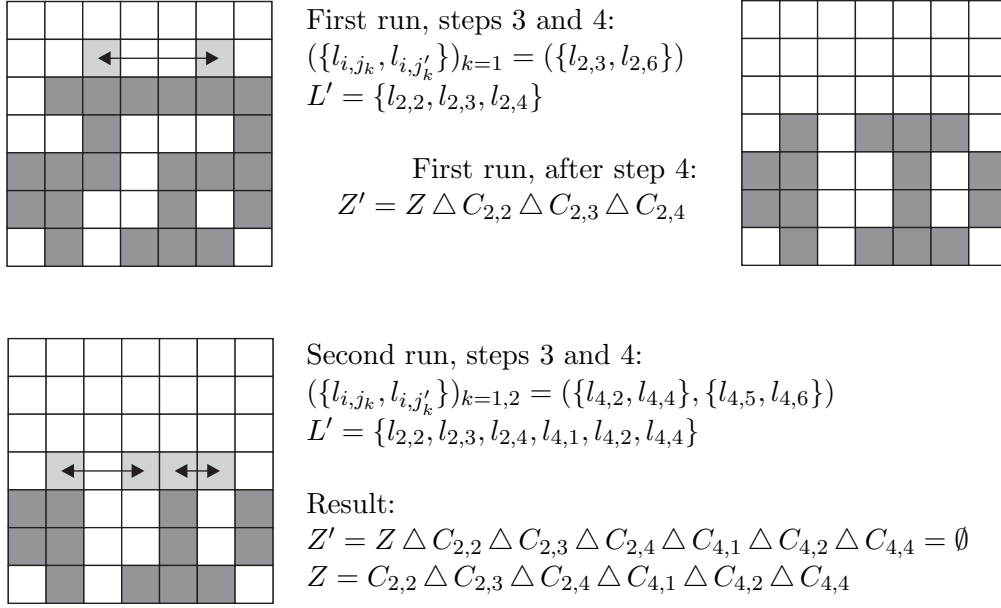


Fig. 7. Application of the algorithm from Theorem 15 to determine the representation of the even S -dominating set from Fig. 6 (c), which we denote with Z , as a linear combination of circles. Shown is the board of lamps with the elements of an even S -dominating set colored grey.

From Proposition 13 and Theorem 15 we immediately have

$$\dim(\ker(A_{S,L})) = \begin{cases} 0 & \text{if } \min(m, n) < 4 \\ (m-3)(n-3) & \text{otherwise.} \end{cases} \quad (3)$$

The rank-nullity theorem from linear algebra states

$$\begin{aligned} \dim(\ker(A_{S,L})) + \dim(\text{rg}(A_{S,L})) &= |L| = mn \\ \dim(\ker(A_{L,S})) + \dim(\text{rg}(A_{L,S})) &= |S| = 3(m+n) - 2 \\ \dim(\text{rg}(A_{S,L})) &= \dim(\text{rg}(A_{L,S})) \end{aligned}$$

which together with (3) gives

$$\dim(\ker(A_{L,S})) = \begin{cases} 3(m+n) - 2 - mn & \text{if } \min(m, n) < 4 \\ 7 & \text{otherwise.} \end{cases} \quad (4)$$

From Proposition 13 we obtain with Corollary 9 that for $\min(m, n) < 4$ all initial light patterns are solvable. The ratio of solvable patterns to all possible patterns is

$$\frac{2^{|S| - \dim(\ker(A_{L,S}))}}{2^{|L|}} = \begin{cases} 1 & \text{if } \min(m, n) < 4 \\ 2^{-mn-9+3(m+n)} & \text{otherwise.} \end{cases}$$

Now that we have revealed the structure of the even S -dominating sets of $G_{m,n}$

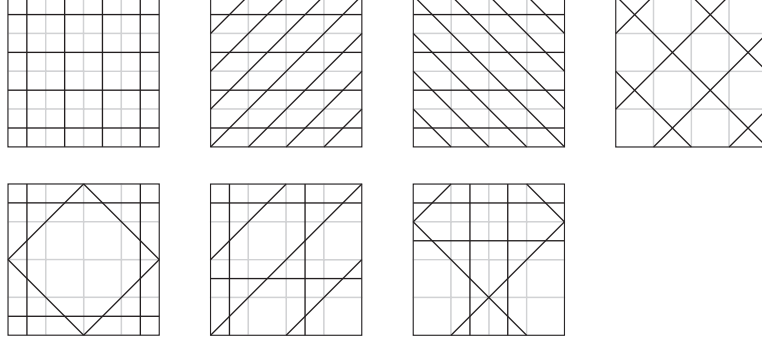


Fig. 8. A basis for the even L -dominating sets of $G_{4,4}$.

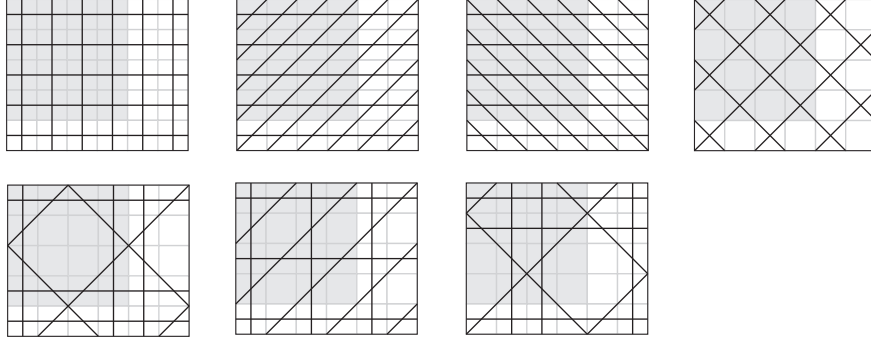


Fig. 9. A basis for the even L -dominating sets of $G_{5,6}$, obtained by tiling the 5×6 board with each of the building blocks from Fig. 8.

we discuss in the following the even L -dominating sets, where we consider only the case $\min(m, n) \geq 4$. Then, as can be seen from (4), their number does not depend on the size of the board of lamps — there exist $2^7 = 128$ even L -dominating sets. A basis of $\ker(A_{L,S})$ for $G_{4,4}$ is illustrated in Fig. 8, all other even L -dominating sets are calculated from the basis as linear combinations. An even L -dominating set as a subset of switches is visualized by drawing lines across the board of lamps at the respective rows, columns, diagonals and antidiagonals. Since each lamp is subject to an even number of switches from the even L -dominating set, an even number of lines intersect in each square.

A basis of $\ker(A_{L,S})$ for $G_{m,n}$ can be constructed from the basis of $\ker(A_{L,S})$ for $G_{4,4}$ by tiling the $m \times n$ board of lamps beginning in a fixed corner with each of the 4×4 building blocks from Fig. 8. This technique is shown in Fig. 9 for $G_{5,6}$.

In addition to the basis of $\ker(A_{L,S})$ in terms of a vector space we want to give a basis of $\ker(A_{L,S})$ in a *geometrical* sense. All even L -dominating sets of $G_{m,n}$ can be obtained by tiling the plane with one of the twelve 4×4 building blocks from Fig. 10 and two operations: shifting an $m \times n$ sized rectangular window over the plane, and inversion (take exactly the switches that are not element of a given even L -dominating set).

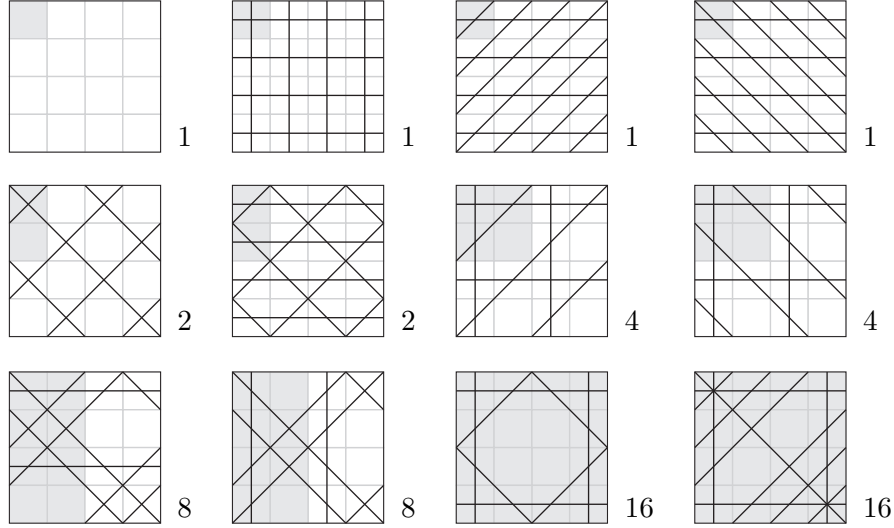


Fig. 10. A geometrical basis for the even L -dominating sets of $G_{m,n}$. The number to the right of each building block indicates how many different even L -dominating sets can be obtained by shifting an $m \times n$ sized rectangular window over the plane tiled with this building block.

These conclusions allow us to solve the geometrical problem stated in the introduction. There are 24 different line patterns on an infinite chessboard that cause an even number of lines to intersect in each square. They are given by the tilings of the plane with each of the twelve building blocks from Fig. 10 and the corresponding inverted tilings. This number reduces to 16 if line patterns that are mapped onto one another by rotation are not counted separately, and to 15 if rotation and reflection are taken into account.

4 Minimum Solutions

The variables $G_{m,n}$, V , E , S , L and A are used as standing notations with respect to the definitions from the previous section. Let U be a finite set. For a vector $x : U \rightarrow \{0, 1\}$ we denote with $w(x)$ the *Hamming weight* of x , defined by $w(x) := \sum_{u \in U} x(u) = |\phi_U(x)|$.

Suppose, an initial light pattern for our Lights Out variant proved to be solvable, e.g. by applying Theorem 8 and Theorem 15. A solution can be found by solving (2) for x , and the set of all solutions is given by $x + \ker(A_{L,S})$. We are concerned with finding a solution that requires only a minimal number of switching operations, we therefore want to determine

$$\min_{x_0 \in \ker(A_{L,S})} w(x + x_0).$$

In the following we formulate criteria to abort the search for a minimum

solution prematurely without calculating the entire solution set explicitly.

Recall, that for the special case $\min(m, n) \geq 4$ the cardinality of $\ker(A_{L,S})$ is independent from the size of the board of lamps and a basis for $\ker(A_{L,S})$ can be constructed easily (see Fig. 8, 9 and 10).

Proposition 16 *Let $z_0 : L \rightarrow \{0, 1\}$ be an initial light pattern and x a solution of (2). Then*

$$\left\lceil \frac{w(z_0)}{\max(m, n)} \right\rceil \leq \min_{x_0 \in \ker(A_{L,S})} w(x + x_0) \leq \left\lfloor \frac{1}{2}(3(m + n) - 2) \right\rfloor. \quad (5)$$

PROOF. The left inequality holds, since with each switching operation at most $\max(m, n)$ lamps are turned off. The right inequality follows from the fact that

$$\underbrace{(1, 1, \dots, 1)}_{|S|=3(m+n)-2}^T$$

is an element of $\ker(A_{L,S})$. If x uses more than half of the switches, then $x + (1, 1, \dots, 1)^T$ is a solution of (2) that uses less than half of the switches. $\square$

Proposition 17 *Let x be a solution of (2) with*

$$w(x) \leq \left\lfloor \frac{1}{2} \left(\min_{x_0 \in \ker(A_{L,S}) \setminus 0} w(x_0) \right) \right\rfloor.$$

Then the equality

$$\min_{x_0 \in \ker(A_{L,S})} w(x + x_0) = w(x)$$

holds.

PROOF. A vector with weight greater or equal to $\min_{x_0 \in \ker(A_{L,S}) \setminus 0} w(x_0)$ added to x gives a vector with weight greater or equal to $w(x)$. $\square$

The application of Proposition 17 requires knowledge about the cardinality of the smallest even L -dominating set of $G_{m,n}$ (except the empty set). Surprisingly, an explicit expression in terms of m and n for the cardinality of the smallest even L -dominating set exists, and even more than that.

For each $k \in \{0, 1, \dots, 3(m + n) - 2\}$ let $h_{m,n}(k)$ denote the number of even L -dominating sets of $G_{m,n}$ with cardinality k , i.e. $h_{m,n}$ is the absolute frequency distribution of the cardinality of the even L -dominating sets.

Table 1

Frequency distribution of the cardinality of the even L -dominating sets of $G_{m,n}$ ($\min(m, n) \geq 4$).

$h_{m,n}(k), k \leq \frac{1}{2}(3(m+n)-2)$																									
m	n																								
$m \pmod{4}$	$n \pmod{4}$																								
		$k=0$	$k=m+n-4$	$k=m+n-3$	$k=m+n-2$	$k=m+n-1$	$k=m+n$	$k=m+n+1$	$k=m+n+2$	$k=1/2(3m+2n-7)$	$k=1/2(3m+2n-5)$	$k=1/2(3m+2n-4)$	$k=1/2(3m+2n-3)$	$k=1/2(3m+2n-2)$	$k=1/2(3m+2n-1)$	$k=1/2(3m+2n)$	$k=1/2(3m+2n+1)$	$k=1/2(2m+3n-5)$	$k=1/2(2m+3n-4)$	$k=1/2(2m+3n-3)$	$k=1/2(2m+3n-2)$	$k=1/2(2m+3n-1)$	$k=1/2(2m+3n)$	$k=1/2(2m+3n+1)$	$k=1/2(2m+3n+3)$
0	0	1	0	0	2	10	15	0	0	0	0	2	0	4	0	2	0	0	2	0	4	0	2	0	0
0	1	1	0	0	0	14	13	0	0	0	0	0	0	8	0	0	0	0	0	4	0	4	0	0	0
0	2	1	0	0	3	10	13	0	1	0	0	2	0	4	0	2	0	0	2	0	4	0	2	0	0
0	3	1	0	0	2	12	11	2	0	0	0	4	0	0	0	4	0	1	0	4	0	2	0	0	1
1	1	1	0	0	7	0	20	0	0	0	2	0	0	0	6	0	0	2	0	0	0	6	0	0	0
1	2	1	0	2	0	10	13	2	0	1	0	0	2	0	4	0	1	0	4	0	0	0	4	0	0
1	3	1	0	0	7	0	20	0	0	0	2	0	0	0	6	0	0	0	0	6	0	0	0	2	0
2	2	1	1	0	0	14	8	4	0	0	0	2	0	4	0	2	0	0	2	0	4	0	2	0	0
2	3	1	0	0	2	12	11	2	0	0	0	0	0	8	0	0	0	0	0	4	0	4	0	0	0
3	3	1	0	0	3	12	8	4	0	0	0	0	6	0	0	0	2	0	0	6	0	0	0	2	0

Theorem 18 Let $\min(m, n) \geq 4$.

Then $h_{m,n}$ is given by the symmetry relations

$$h_{m,n} = h_{n,m} \quad (6)$$

$$\bigwedge_{k \in \{0,1,\dots,3(m+n)-2\}} h_{m,n}(k) = h_{m,n}(3(m+n)-2-k) \quad (7)$$

and Table 1, where columns with the same label (for a certain m and n) have to be added up.

PROOF. Relation (6) holds, since $G_{m,n}$ and $G_{n,m}$ are isomorphic. Relation (7) follows from the fact, that S is an even L -dominating set (cp. the proof for Proposition 16). Each row of Table 1 can be computed by counting the number of switches in the even L -dominating sets constructed from the geometrical basis in Fig. 10 (using a computer program simplifies this work considerably). $\square$

Example. To examine $G_{4,4}$ consider the first row of Table 1. The column labels for nonzero entries are $k = 0, 6, 7, 8, 8, 9, 10, 8, 9, 10, 10, 11, 12, 10, 11, 12, 10, 11$. The columns with $k = 12$ are dropped, since $k = 12 \not\leq \frac{1}{2}(3(4+4)-2) = 11$ holds. Adding up the entries in columns with the same label yields Table 2. From (7) we eventually obtain $h_{4,4}(k)$ for $11 < k \leq 22$. The complete frequency distribution is depicted in Fig. 11.

Table 2

Frequency distribution of the cardinality of the even L -dominating sets of $G_{4,4}$.

	0	6	7	8	9	10	11	otherwise
$h_{4,4}(k), k \leq 11$	k	k	k	k	k	k	k	
	1	2	10	19	8	10	28	0

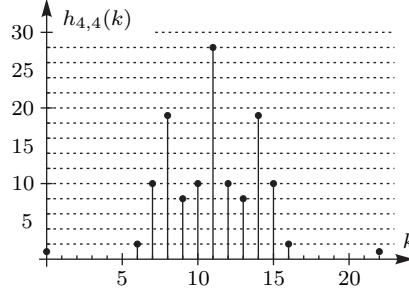
Fig. 11. Frequency distribution of the cardinality of the even L -dominating sets of $G_{4,4}$.

Table 3

Minimal cardinality of the even L -dominating sets of $G_{m,n}$ ($\min(m, n) \geq 4$).

$\min_{x_0 \in \ker(A_{L,S}) \setminus 0} w(x_0)$		$n \pmod{4}$			
		0	1	2	3
$m \pmod{4}$	0	$m+n-2$	$m+n-1$	$m+n-2$	$m+n-2$
	1	$m+n-1$	$m+n-2$	$m+n-3$	$m+n-2$
	2	$m+n-2$	$m+n-3$	$m+n-4$	$m+n-2$
	3	$m+n-2$	$m+n-2$	$m+n-2$	$m+n-2$

Corollary 19 *It is*

$$\min_{x_0 \in \ker(A_{L,S}) \setminus 0} w(x_0) = \begin{cases} 2 & \text{if } \min(m, n) = 1 \\ 3 & \text{if } \min(m, n) = 2 \\ m+n-2 & \text{if } \min(m, n) \geq 3 \text{ and} \\ & (m \in 4\mathbb{N} + 3 \text{ or } n \in 4\mathbb{N} + 3 \text{ or } m, n \in 4\mathbb{N}) \\ 4(\lfloor \frac{m}{4} \rfloor + \lfloor \frac{n}{4} \rfloor) & \text{otherwise.} \end{cases}$$

PROOF. For $\min(m, n) = 1$, $\min(m, n) = 2$ and $\min(m, n) = 3$ even L -dominating sets of $G_{m,n}$ with minimal cardinality 2, 3 and $\max(m, n) + 1 = m + n - 2$, respectively, are depicted in Fig. 12. For $\min(m, n) \geq 4$ we obtain Table 3 as a special case from Table 1. The shaded entries can be written as $4(\lfloor \frac{m}{4} \rfloor + \lfloor \frac{n}{4} \rfloor)$. $\square$

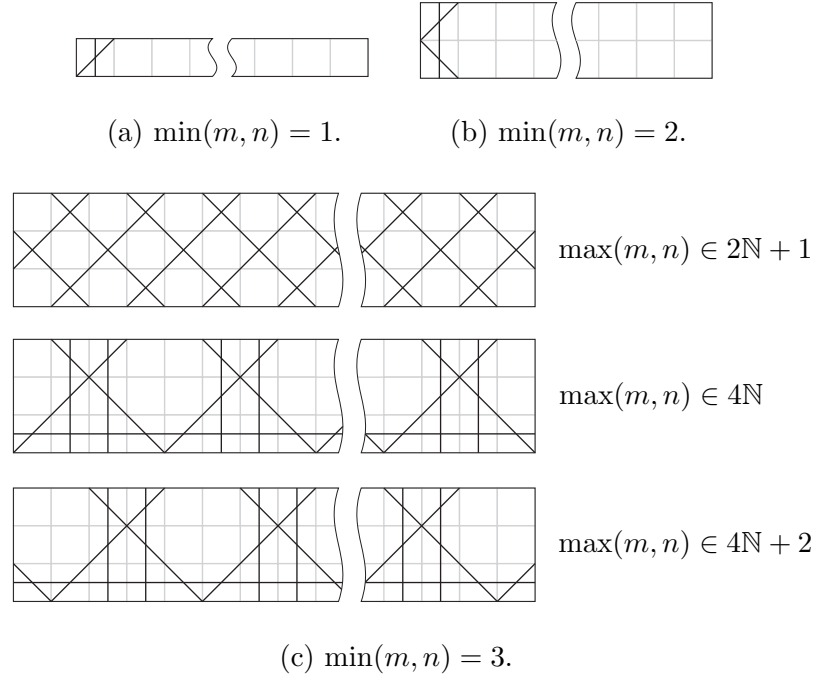


Fig. 12. Even L -dominating sets of $G_{m,n}$ ($\min(m, n) \in \{1, 2, 3\}$) with minimal cardinality.

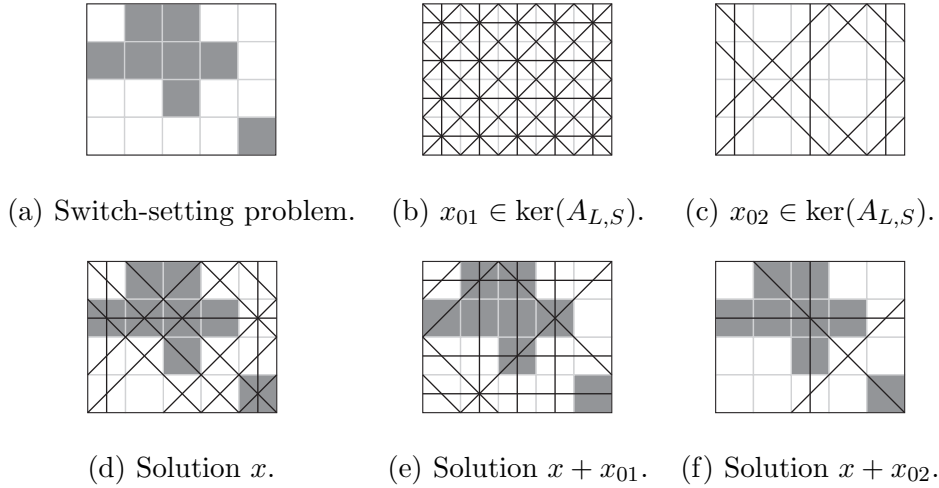


Fig. 13. Search for a minimum solution (lit lamps are colored grey).

Example. Consider the light pattern on the 4×5 board of lamps shown in Fig. 13 (a). According to Theorem 8 this switch-setting problem is solvable if the number of lit lamps on every even S -dominating set from a base of the set of all even S -dominating sets of $G_{4,5}$ is even. This condition is fulfilled, the number of lit lamps on the circles $C_{1,1}$ and $C_{1,2}$ is 4 and 2, respectively. A solution x of (2) with $w(x) = 13$ is depicted in Fig. 13 (d). Two even L -dominating sets corresponding to vectors $x_{01}, x_{02} \in \ker(A_{L,S})$ with $w(x_{01}) = 25$ and $w(x_{02}) = 11$ are shown in Fig. 13 (b) and (c). From Proposition 16 we get $2 \leq \min_{x_0 \in \ker(A_{L,S})} w(x + x_0) \leq 12$, and Corollary 19 gives

Table 4

Solutions for zero-one integer linear programming problem.

Ω		n											
		1	2	3	4	5	6	7	8	9	10	11	12
m	1	1	1	2	2	3	3	4	4	5	5	6	6
	2	1	2	3	4	5	5	6	—	—	—	—	—
	3	2	3	5	6	9	10	12	12	14	15	17	18
	4	2	4	6	8	10	11	13	14	15	17	—	20
	5	3	5	9	10	12	12	14	15	17	—	21	22
	6	3	5	10	11	12	14	15	17	19	20	22	23
	7	4	6	12	13	14	15	17	19	22	23	24	25
	8	4	—	12	14	15	17	19	22	24	24	—	—
	9	5	—	14	15	17	19	22	24	25	25	—	—
	10	5	—	15	17	—	20	23	24	25	26	—	29
	11	6	—	17	—	21	22	24	—	—	—	29	—
	12	6	—	18	20	22	23	25	—	—	29	—	—

$\min_{x_0 \in \ker(A_{L,S}) \setminus 0} w(x_0) = 8$. A second solution is $x + x_{01}$ with $w(x + x_{01}) = 12$ (see Fig. 13 (e)). According to Proposition 17, $x + x_{02}$ with $w(x + x_{02}) = 4$ is a minimum solution (see Fig. 13 (f)).

To tighten the upper bound in (5) further we consider the relation

$$\min_{x_0 \in \ker(A_{L,S})} w(x + x_0) \leq \underbrace{\max_{x' \in \{0,1\}^S} \left(\min_{x_0 \in \ker(A_{L,S})} w(x' + x_0) \right)}_{=: \Omega} \leq \left\lfloor \frac{1}{2}(3(m+n) - 2) \right\rfloor.$$

The max-min problem to determine Ω can be written equivalently as the following zero-one integer linear programming problem.

Problem Find an $x' \in \{0,1\}^S$ to maximize the linear objective function

$$w(x') = \sum_{s \in S} x'(s),$$

subject to the linear inequality constraints

$$\bigwedge_{x_0 \in \ker(A_{L,S})} x_0^T x' \leq \left\lfloor \frac{w(x_0)}{2} \right\rfloor.$$

Computer experiments for $m, n \leq 12$ with the integer programming module of the NAG Fortran 90 library give the results in Table 4, where problems for which no solution is known are marked with a dash. Note, that for $(m, n) = (8, 9)$ and $(m, n) = (9, 8)$ it is $\Omega = \lfloor \frac{1}{2}(3(m+n) - 2) \rfloor$.

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